

3D ORGAN LOCATION PREDICTION PIPELINE: END-TO-END ARCHITECTURE

Real-Time Patient Body Surface Point Cloud Sensing + PointNet++ Multi-Scale Encoder + Target-Query Transformer Decoder

Rigorous Physical Normalization ($S_{global} = 500.0$ mm) | Scientific Dataset V3 Frozen Protocol (1,668 Subjects, 104 Canonical Targets)

FUNDAMENTAL CLINICAL PARADIGM: ZERO CT AT REAL-TIME INFERENCE

- OFFLINE TRAINING ONLY: Full-body diagnostic CT is used strictly offline to generate ground-truth organ centroids & extract exterior skin points.
- REAL-TIME CLINICAL INFERENCE: Live non-contact optical 3D surface scanner captures patient's exterior skin surface ONLY (4096 points, 0 mSv radiation).
- ZERO INTERNAL IMAGES AT TEST TIME: CT intensities, HU values, and organ segmentation masks are NEVER needed during live operation!

TRACK 1: OFFLINE TRAINING

CT Ground Truth Annotation (Offline Only)



Diagnostic Full-Body CT Volume
(512 x 512 x z Voxels, Offline)

1. Automated 3D Segmentation

TotalSegmentator / AMOS AI
Extracts 3D masks for 104 targets
Ground-Truth Centroid:
 $C_{GT} = \frac{1}{|V|} \sum_{V \in \mathbb{R}^{104 \times 3}} V$

2. External Surface Extraction

Skin mask threshold (HU > -300)
Marching Cubes isosurface extraction
Yields dense outer skin surface
Zero internal tissue masks retained

3. Farthest Point Sampling

Downsample skin mesh to
 $N = 4096$ surface points
Uniform spatial coverage over
patient exterior skin boundary

4. Centering & Normalization

$C_{body} = \frac{1}{N}(\min P + \max P)$
 $P_{norm} = (P - C_{body})/500.0$ mm
 $C_{norm}^{GT} = (C_{GT} - C_{body})/500.0$ mm

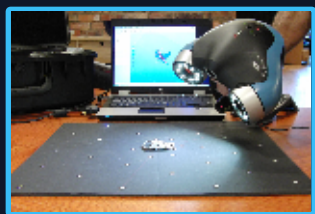
OFFLINE SUPERVISED PAIRS

Pair: $(P_{norm}, C_{norm}^{GT})$

ALL CT VOXELS & MASKS
PERMANENTLY DISCARDED!
Model never accesses CT at runtime.

TRACK 2: REAL-TIME INFERENCE

Radiation-Free Live Optical Body Scan



Live Optical 3D Surface Scanner
(Structured Light / Stereoscopic / LiDAR / SGRT)
E.g. AlignRT, VisionRT, RealSense, Kinect

1. Live Non-Contact Surface Capture

- Captures exterior skin surface geometry
- Dense 3D point cloud: 50,000 - 200,000 pts
- Stream Rate: 30 - 60 Hz (Continuous Live)
- Radiation: 0 mSv (Safe optical visible light)

2. Real-Time Preprocessing (< 5 ms)

- Treatment couch / bed plane removal
- Torso Region-of-Interest (ROI) crop
- Statistical outlier & sensor noise filter
- Uniform voxel grid downsampling to 4096 pts

3. Centering & Physical Scale

Bounding Box Center:
 $C_{body} = \frac{1}{2}(\min(P_{box}) + \max(P_{box}))$
 $P_{centered} = P_{live} - C_{body}$
 $P_{norm} = P_{centered}/500.0$ mm

LIVE MODEL INPUT TENSOR

$X_{in} \in \mathbb{R}^{B \times 4096 \times 3}$ (Exterior Skin Only)

- ZERO CT Intensities Needed
- ZERO Organ Masks Needed
- Pure Geometric 3D Surface Input

REAL-TIME LATENCY BUDGET

1. Optical Surface Sensing: 8 - 12 ms
 2. Real-Time Preprocessing: 3 - 5 ms
 3. GPU Model Forward Pass: 12 - 15 ms
 4. Denormalization & Render: 1 - 2 ms
- TOTAL LATENCY: ~25 - 34 ms (30+ FPS)

TRACK 3: TARGET-QUERY TRANSFORMER MODEL ARCHITECTURE

PointNet++ Multi-Scale Surface Encoder + 4-Layer Cross-Attention Decoder

STAGE 1: POINTNET++ ENCODER

Surface Point Cloud Input

$P_{norm} \in \mathbb{R}^{B \times 4096 \times 3}$

SA1: Fine Surface Geometry

$N_1 = 1024$ | Radius $r_1 = 0.2$ (100 mm) | $K = 32$
MLP: [64, 64, 128] $\rightarrow$ (B, 1024, 128)

SA2: Mid-Scale Surface Tokens

$N_2 = 256$ | Radius $r_2 = 0.4$ (200 mm) | $K = 32$
MLP: [128, 128, 256] $\rightarrow$ (B, 256, 256) [MID]

SA3: Coarse Torso Tokens

$N_3 = 64$ | Radius $r_3 = 0.8$ (400 mm) | $K = 32$
MLP: [256, 256, 512] $\rightarrow$ (B, 64, 512) [COARSE]

SA4: Global Pooled Vector

All Points Pooling | MLP: [512, 512, 1024]
Global Torso Context: (B, 1024)

Surface Memory Assembler

- Proj $L_2(256 \rightarrow 256)$ + ScaleEmbo
 - Proj $L_2(512 \rightarrow 256)$ + ScaleEmbo
 - Concatenate: $256 + 64 = 320$ Surface Tokens
 - Continuous 3D Positional Encoding: $PE(X_{xyz})$
- $M_{tokens} \in \mathbb{R}^{B \times 320 \times 256}$

Surface Memory
Tokens (K, V)
(320 Tokens)

STAGE 2: TRANSFORMER DECODER

Target Anatomical Queries

- 104 Learned Organ Embeddings ($E_k \in \mathbb{R}^{1 \times 256}$)
 - Canonical Atlas Centroid ($C_{atlas} \in \mathbb{R}^{104 \times 3}$)
 - Atlas Positional Encoding: $PE(C_{atlas})$
- $Q_0 = E_k + \text{AtlasPE}(C_{atlas}, k) \in \mathbb{R}^{B \times 104 \times 256}$

4x Transformer Decoder Layers

Multi-Head Cross-Attention (8 Heads)

Queries (Q) attend to 320 Surface Tokens (K, V)
Geometric Bias: Relative vector ($X_{surf} - C_{atlas}$)

Residual & LayerNorm

Residual Connection + LayerNorm(256)

Feed-Forward Network (FFN)

Linear(256 $\rightarrow$ 1024) $\rightarrow$ ReLU $\rightarrow$ Linear(1024 $\rightarrow$ 256)
Residual Connection + LayerNorm(256)

STAGE 3: COORDINATE RESIDUAL HEAD

Linear(256 $\rightarrow$ 128) $\rightarrow$ LayerNorm $\rightarrow$ ReLU $\rightarrow$ Linear(128 $\rightarrow$ 3)

Predicts Centroid Residual Offset: $\Delta P_k \in \mathbb{R}^3$

Normalized Output Formulation:

$\hat{C}_{norm, k} = C_{atlas, k} + \Delta P_k$

